

Week 11-Count data: Negative Binomial Response

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1 Negative Binomial Response (section 3.3, Faraway)

The response variable, skips, is a count of how many solder skips appeared to a visual inspection.

1.1 Poisson regression

We start with a Poisson regression:

```
#Library(faraway)
#?solder
data(solder, package="faraway")
modp <- glm(skips ~ . , family=poisson, data=solder)
summary(modp)
```

```
##
## Call:
## glm(formula = skips ~ ., family = poisson, data = solder)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.10042    0.09267 -11.874 < 2e-16 ***
## OpeningM      0.57161    0.05707  10.017 < 2e-16 ***
## OpeningS      1.81475    0.05044  35.980 < 2e-16 ***
## SolderThin    0.84682    0.03327  25.453 < 2e-16 ***
## MaskA3        0.51315    0.07098   7.230 4.83e-13 ***
## MaskA6        1.81103    0.06609  27.404 < 2e-16 ***
## MaskB3        1.20225    0.06697  17.953 < 2e-16 ***
## MaskB6        1.86648    0.06310  29.580 < 2e-16 ***
## PadTypeD6     -0.40328    0.06028  -6.690 2.24e-11 ***
## PadTypeD7     -0.08979    0.05521  -1.626 0.103852
## PadTypeL4      0.22460    0.05117   4.389 1.14e-05 ***
## PadTypeL6     -0.70633    0.06637 -10.642 < 2e-16 ***
## PadTypeL7     -0.48260    0.06176  -7.814 5.52e-15 ***
## PadTypeL8     -0.30778    0.05862  -5.251 1.51e-07 ***
## PadTypeL9     -0.63793    0.06489  -9.831 < 2e-16 ***
## PadTypeW4     -0.19021    0.05671  -3.354 0.000796 ***
## PadTypeW9     -1.56252    0.09165 -17.048 < 2e-16 ***
## Panel         0.14670    0.01745   8.405 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 8788.2  on 899  degrees of freedom
## Residual deviance: 1829.0  on 882  degrees of freedom
## AIC: 3967.6
##
## Number of Fisher Scoring iterations: 5
```

```
c(deviance(modp), df.residual(modp))
```

```
## [1] 1829.002  882.000
```

We see that the full model has a residual deviance of 1829 on 882 degrees of freedom. This is not a good fit.

```
modp2 <- glm(skips ~ (Opening +Solder + Mask + PadType + Panel)^2 , family=poisson, data=sold
er)
c(deviance(modp2), df.residual(modp2))
```

```
## [1] 1068.817  882.000
```

The Goodness of fit test:

```
pchisq(deviance(modp2),df.residual(modp2),lower=FALSE)
```

```
## [1] 1.130696e-10
```

Still not enough to conclude that the model fits.

We could try adding more interactions but that would make interpretation increasingly difficult. A check for outliers reveals no problem.

1.2 Negative binomial

An alternative model for counts is the negative binomial. Set $k = 1$ (geometric distribution)

```
library(MASS) #negative.binomial(k) function
modn <- glm(skips ~ ., negative.binomial(1), solder)
summary(modn)
```

```
##
## Call:
## glm(formula = skips ~ ., family = negative.binomial(1), data = solder)
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.69933    0.16334 -10.404 < 2e-16 ***
## OpeningM     0.50854    0.08797   5.781 1.03e-08 ***
## OpeningS     1.99966    0.08069  24.783 < 2e-16 ***
## SolderThin    1.04894    0.06364  16.483 < 2e-16 ***
## MaskA3        0.65710    0.10463   6.280 5.29e-10 ***
## MaskA6        2.52649    0.12123  20.841 < 2e-16 ***
## MaskB3        1.27261    0.10780  11.806 < 2e-16 ***
## MaskB6        2.08026    0.10482  19.845 < 2e-16 ***
## PadTypeD6     -0.46118    0.13788  -3.345 0.000859 ***
## PadTypeD7      0.01608    0.13350   0.120 0.904185
## PadTypeL4      0.46883    0.13046   3.594 0.000344 ***
## PadTypeL6     -0.47115    0.13799  -3.414 0.000669 ***
## PadTypeL7     -0.29494    0.13620  -2.165 0.030618 *
## PadTypeL8     -0.08493    0.13432  -0.632 0.527372
## PadTypeL9     -0.52125    0.13854  -3.763 0.000179 ***
## PadTypeW4     -0.14250    0.13481  -1.057 0.290781
## PadTypeW9     -1.48361    0.15317  -9.686 < 2e-16 ***
## Panel         0.16932    0.03811   4.443 1.00e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(1) family taken to be 0.5560534)
##
##      Null deviance: 1743.01  on 899  degrees of freedom
## Residual deviance:  558.67  on 882  degrees of freedom
## AIC: 3883.7
##
## Number of Fisher Scoring iterations: 9
```

We could experiment with different values of k , but there is a more direct way of achieving this by allowing the parameter k to vary and be estimated using maximum likelihood.

$$Y|E \sim \text{Poisson}(\mu E), \quad \theta E \sim \text{gamma}(\theta)$$

$$E(E) = 1, \quad \text{Var}(E) = 1/\theta$$

The posterior: $\text{Bin}(k = \theta, p = \theta/(\mu + \theta))$

$$f_Y(y; \theta, \mu) = \frac{\Gamma(\theta + y)}{\Gamma(\theta)y!} \frac{\mu^y \theta^\theta}{(\mu + \theta)^{\theta+y}}$$

$$E(Y) = \mu, \quad \text{Var}(Y) = \mu + \mu^2/\theta$$

```
modn <- glm.nb(skips ~ .,solder) #glm.nb {MASS}
summary(modn)
```

```
##
## Call:
## glm.nb(formula = skips ~ ., data = solder, init.theta = 4.397157245,
##       link = log)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.42245    0.14274  -9.965  < 2e-16 ***
## OpeningM      0.50294    0.07976   6.306 2.87e-10 ***
## OpeningS      1.91317    0.07152  26.750 < 2e-16 ***
## SolderThin     0.93932    0.05362  17.517 < 2e-16 ***
## MaskA3         0.58981    0.09651   6.112 9.87e-10 ***
## MaskA6         2.26734    0.10182  22.269 < 2e-16 ***
## MaskB3         1.21101    0.09637  12.566 < 2e-16 ***
## MaskB6         1.99037    0.09223  21.580 < 2e-16 ***
## PadTypeD6     -0.46592    0.11238  -4.146 3.38e-05 ***
## PadTypeD7     -0.03315    0.10673  -0.311 0.756114
## PadTypeL4      0.38268    0.10265   3.728 0.000193 ***
## PadTypeL6     -0.57844    0.11413  -5.068 4.01e-07 ***
## PadTypeL7     -0.36656    0.11094  -3.304 0.000953 ***
## PadTypeL8     -0.15890    0.10821  -1.468 0.141986
## PadTypeL9     -0.56600    0.11393  -4.968 6.77e-07 ***
## PadTypeW4     -0.20044    0.10873  -1.844 0.065255 .
## PadTypeW9     -1.56460    0.13621 -11.486 < 2e-16 ***
## Panel         0.16369    0.03139   5.214 1.85e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(4.3972) family taken to be 1)
##
##      Null deviance: 4043.3  on 899  degrees of freedom
## Residual deviance: 1008.3  on 882  degrees of freedom
## AIC: 3683.3
##
## Number of Fisher Scoring iterations: 1
##
##              Theta:  4.397
##             Std. Err.: 0.495
##
## 2 x log-likelihood: -3645.309
```

We see that $\hat{k} = 4.397$